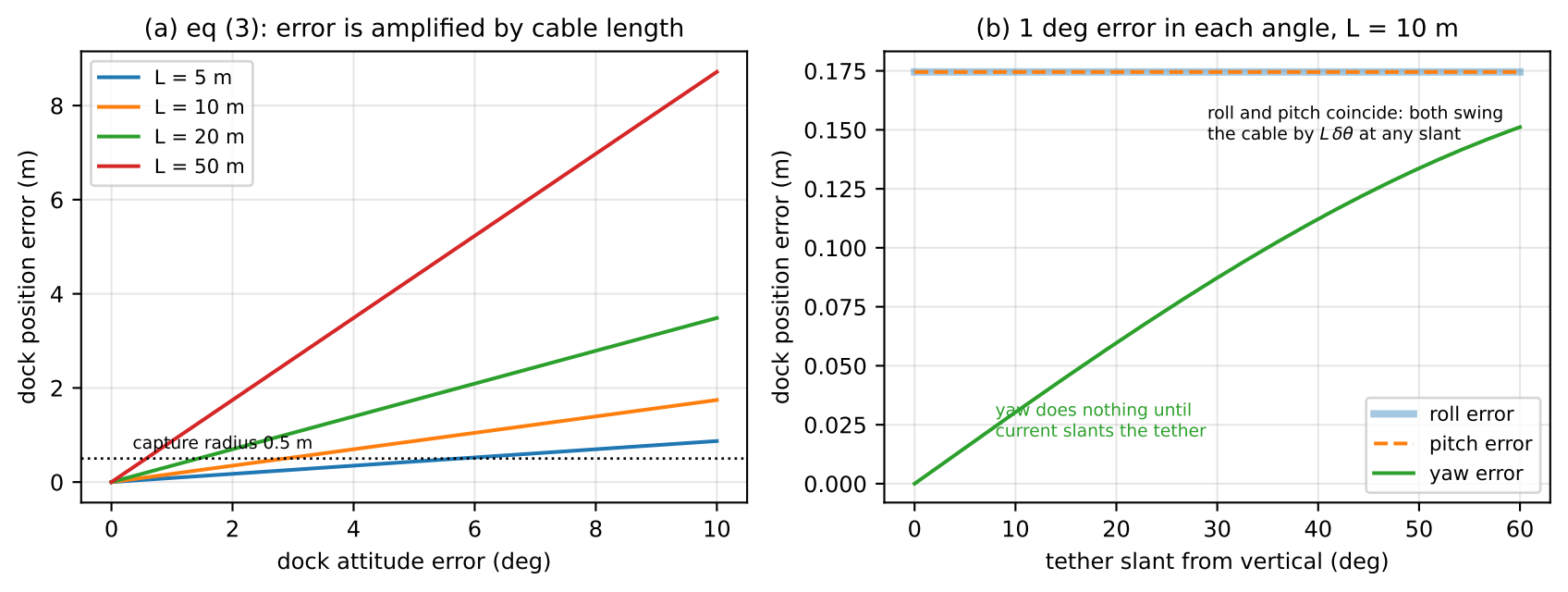
(a) eq (3): error is amplified by cable length
L = 5 m
L = 10 m
L = 20 m
L = 50 m
capture radius 0.5 m
dock position error (m)
dock attitude error (deg)
0
2
4
6
8
10
(b) 1 deg error in each angle, L = 10 m
roll and pitch coincide: both swing
the cable by $L\delta\theta$ at any slant
yaw does nothing until
current slants the tether
roll error
pitch error
yaw error
dock position error (m)
tether slant from vertical (deg)
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0
10
20
30
40
50
60